Personalizing POI Descriptions with LLMs: A Study of User Perceptions and Informational Value

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Abstract

Personalized content is essential for engaging users in travel planning, yet most Points of Interest (POI) descriptions remain generic and static. This study examines whether large language models (LLMs) can effectively personalize POI descriptions to enhance user engagement and perceived information quality. A within-subjects study was conducted with 51 participants who evaluated both original and LLM-adapted descriptions across 10 diverse POIs, including hotels, restaurants, museums, and gardens. Participants rated descriptions for significance, trustworthiness, and clarity. Results indicate that AI-personalized descriptions significantly improved perceived clarity ($p = 0.0218$) and showed marginal improvements in trustworthiness ($p = 0.050$) compared to manual descriptions. AI adaptations shifted user responses from neutral assessments toward stronger positive endorsements while maintaining engagement levels similar to carefully crafted manual content. Participants also reported high comfort with AI-generated travel content (mean = 3.94 on a 5-point scale). These findings suggest that LLM-based personalization provides a scalable approach to improve tourism content relevance and user engagement without compromising credibility or information quality.

CCS Concepts

• **Human-centered computing** → **User centered design**; *User studies.*

Keywords

poi adaptation, LLM-generated content, ai-driven descriptions, content personalization

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1 Introduction

Personalization has become a key factor in improving user engagement in various digital experiences, including travel and tourism. As travellers seek more relevant and tailored recommendations, AI advancements, particularly large language models (LLM), presents new opportunities to adapt content to align with individual preferences dynamically. One such application is personalized descriptions of Points of Interest (POIs), such as hotels, restaurants, museums, and gardens. Although traditional POI descriptions are often static and generalized, AI-driven personalization can refine these descriptions to match user-specific interests, past experiences, and contextual relevance, enhancing both engagement and comprehension.

Much prior research has focused on POI recommendation and ranking systems, where AI models suggest locations based on user preferences, behaviors, and contextual factors. However, less attention has been given to how the content of each POI itself can be adapted to better engage users. When individuals access POI descriptions, the same information is typically presented to all users, despite variations in their backgrounds, interests, and motivations to visit a place. For example, a museum description could emphasize historical significance for a history enthusiast while focusing on interactive exhibits for a family traveler. Generative AI and LLMs provide a unique opportunity to dynamically tailor content without altering its factual meaning, ensuring that the description remains relevant and appealing to different audiences.

To explore this potential, a prototype system was developed that allows users to enter their preferences and interests and then dynamically presents two versions of each POI description: the original and an LLM-adapted version. Participants were asked to evaluate the descriptions based on alignment with their preferences, perceived appeal, and familiarity with the location. The key objective is to assess whether AI-generated adaptations are acceptable and meaningful, effectively convey the original value of POI, and improve user engagement.

This study contributes to the growing field of AI-powered content personalization in tourism by investigating the role of LLMs in dynamic adaptation of textual content. By analyzing participant feedback, this study aims to provide insight into how AI-generated descriptions can improve engagement while preserving accuracy and authenticity.

2 Related work

Prior work on tourism and location-based personalization has largely focused on point-of-interest (POI) recommendation, where systems predict or rank places that match a user profile and context. Surveys of travel and POI recommenders highlight the importance of heterogeneous signals such as mobility patterns, time, geography, and user attributes, as well as evaluation challenges that arise from contextual variability and data sparsity [6, 10]. Recent work in the UMAP community also emphasizes that user preferences can differ before and after a visit, which complicates modeling and suggests that preference signals and content expectations are not static [2].

With the rise of large language models (LLMs), research has expanded from representation learning and ranking to interactive

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and generative paradigms for personalization. In the POI domain, recent overviews and systems propose LLM-based pipelines that augment data, unify encoding, and support conversational interaction [8, 9]. Empirical studies and methods also explore how LLMs can be used for next-POI recommendation and for generating user profiles that act as conditioning context [5, 11]. In recommender systems more broadly, LLMs are increasingly studied as general-purpose personalization components that can support explanation, dialogue, and reasoning, but they also raise concerns about faithfulness and evaluation, which relate directly to perceived trust and clarity [1, 7, 13].

In parallel, tourism research has begun to evaluate generative AI for content creation and trip planning, including marketing materials and domain-specific models [3, 4, 12]. These works motivate the use of generative models in tourism, but they often study destination promotion or end-to-end planning rather than the personalization of the micro-content that users read when deciding whether a POI is relevant. This gap is addressed by adapting existing POI descriptions to a user-provided interest profile while keeping core facts stable, and by measuring user perceptions of significance, trustworthiness, and clarity in a controlled within-subjects study.

3 Methodology

To evaluate the effectiveness of LLM-based personalization of POI descriptions, a controlled user study was conducted comparing original and AI-adapted descriptions. This section details the study design, POI selection, personalization approach, evaluation procedure, and participant characteristics.

3.1 Study Design

A within-subjects experimental design was employed in which each participant evaluated both original and AI-personalized descriptions for multiple POIs. This design enabled direct comparison between description types while controlling for individual differences in preferences and expectations. The study presented participants with 10 POIs spanning five categories: hotels, restaurants, museums, gardens, and landmarks. Each category included two POIs to ensure diverse representation of travel-related content. POIs were selected to include both internationally recognized locations and lesser-known destinations, providing variety in familiarity levels.

3.2 Description Personalization

For each POI, two description versions were prepared. The original description was sourced from official tourism websites and travel platforms, representing standard, non-personalized content. The AI-adapted version was generated using a large language model (GPT-4) that received the original description along with each participant's stated interests, hobbies, and travel preferences as input. The LLM was instructed to adapt the description to emphasize aspects most relevant to the user's profile while maintaining factual accuracy and the core information about the POI. For example, a museum description for a participant interested in history would emphasize historical significance and artifacts, while the same museum described for a family traveler would highlight interactive exhibits and educational programs.

3.3 Evaluation Interface

Participants viewed descriptions through a web-based interface that displayed two versions side by side, labeled as Version A and Version B (see Figure 1). The interface did not indicate which version was AI-generated. Each POI was accompanied by a representative image to provide visual context. The presentation order of AI and manual descriptions was randomized across POIs to prevent order bias. Participants could not proceed to the next POI without completing evaluations for the current one, ensuring complete data collection.

3.4 Evaluation Metrics

Participants evaluated each description on three core dimensions using Likert scales:

Significance: "Does the description effectively communicate the significance and offerings of the place?" rated on a 5-point scale from Strongly Disagree to Strongly Agree. This metric assessed whether the description conveyed the value and appeal of the POI.

Trustworthiness: "How trustworthy does this description appear to be?" rated on a 5-point scale from Not at all to Extremely. This metric measured perceived credibility and reliability of the information.

Clarity: "How clear and complete is the information provided?" rated on a 5-point scale from Very Unclear to Very Clear. This metric evaluated information quality and comprehensibility.

After rating both descriptions individually, participants completed comparative evaluations selecting which description was more engaging, more informative, more appealing, and more likely to encourage them to visit. Participants also reported their familiarity with each POI using three categories: visited in person, heard about it, or never heard of it.

3.5 Procedure

The study was conducted in three phases using a secure web platform. First, participants provided informed consent and completed a survey regarding their demographics, travel styles, and interests. This data was used to personalize the AI-generated descriptions. Second, participants evaluated 10 POIs sequentially by reading both description versions and rating them on core metrics and familiarity. This evaluation phase lasted approximately 20 to 30 minutes. Finally, participants provided overall feedback on their experience and their comfort level with AI-generated travel content. All data were collected anonymously to ensure privacy and ethical compliance.

3.6 Participants

A total of 51 participants were recruited through personal networks via email and direct messaging. Participants were required to be at least 18 years old and possess basic English proficiency to ensure effective engagement with POI descriptions. No specific travel experience was mandated, as the study aimed to assess AI-generated descriptions across diverse user segments.

The sample demographics were as follows. Ages ranged from 22 to 49 years (M = 30.94, SD = 6.74). Gender distribution included 34 males (67%) and 17 females (33%). Nationalities were predominantly Indian (n = 28) and German (n = 21), with minimal Chinese and Iranian representation. Marital status comprised 25 married, 22

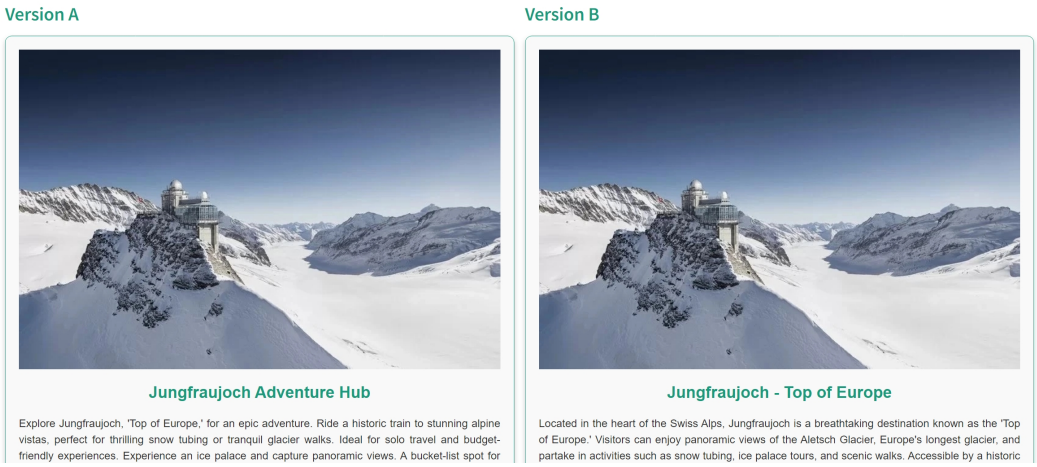

Figure 1: Evaluation interface showing two description versions (Version A and Version B) presented side by side for participant comparison

single, 3 in relationships, and 1 divorced. Professional backgrounds centered on students (n = 22) and engineers (n = 16), with smaller numbers of academics, healthcare professionals, and public service workers.

Travel experience levels showed intermediate (n = 21) and experienced (n = 19) travelers forming the majority, with 9 beginners and 2 experts. This distribution provided a range of perspectives on travel content while maintaining concentration in engaged, technology-familiar demographics.

4 Results

This section presents the findings from the user study evaluating LLM-adapted POI descriptions against original manual descriptions. Participant responses are analyzed across three core metrics: significance, trustworthiness, and clarity, followed by comparative preference evaluations and overall feedback on the adaptive personalization concept. The study collected 510 paired observations (51 participants × 10 POIs) for each metric, enabling robust statistical analysis of user perceptions. Table 1 provides an overview of the rating distributions across all three evaluation dimensions.

4.1 Perceived Significance

Participants rated both description versions on how effectively they communicated the significance and offerings of each POI using a 5-point Likert scale ranging from Strongly Disagree to Strongly Agree. Figure 2 illustrates the distribution of responses for manual and AI-generated descriptions.

The results show that both manual and AI descriptions received predominantly positive ratings. In both conditions, the largest share of responses fell in level 4 (Agree; Manual: 55.1%, AI: 54.1%), with additional responses at the highest level 5 (Strongly Agree; Manual: 12.3%, AI: 16.5%). AI-generated descriptions exhibited a shift toward stronger endorsement, reflected in a 4.2 percentage-point increase in top-box ratings (level 5) and a corresponding reduction in neutral responses (level 3; Manual: 27.1%, AI: 23.1%). A chi-square test on the underlying $N = 510$ observations revealed no statistically significant difference between distributions ($\chi^2(4) = 4.87$, $p = 0.3005$), indicating that AI adaptations maintained comparable perceived significance while trending toward higher extreme-positive ratings.

Table 1: Distribution of User Ratings % for Manual vs. AI Descriptions ($N = 510$)

Metric	Level (1–5)	Manual	AI
Significance	1 (Lowest)	1.4	1.6
	2	4.1	4.7
	3	27.1	23.1
	4	55.1	54.1
	5 (Highest)	12.3	16.5
Trust	1 (Lowest)	1.6	0.8
	2	8.4	6.5
	3	33.3	27.4
	4	46.1	51.2
	5 (Highest)	10.6	14.1
Clarity	1 (Lowest)	0.6	1.0
	2	6.5	4.7
	3	28.8	21.4
	4	51.0	55.7
	5 (Highest)	13.1	17.2

4.2 Trustworthiness

Trustworthiness was assessed on a 5-point scale ranging from Not at all to Extremely. Figure 3 presents the comparative distribution of trust ratings.

AI-generated descriptions received higher trust ratings overall than manual descriptions. At the top end of the scale, the share of "Extremely" trustworthy responses (level 5) increased from 10.6% (manual) to 14.1% (AI), while "Very" trustworthy responses (level

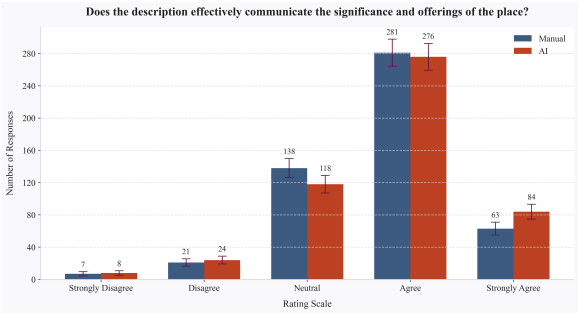

Figure 2: Comparison of perceived significance between manual and AI-generated descriptions. Error bars represent Poisson standard error estimates ($\sqrt{n}$).

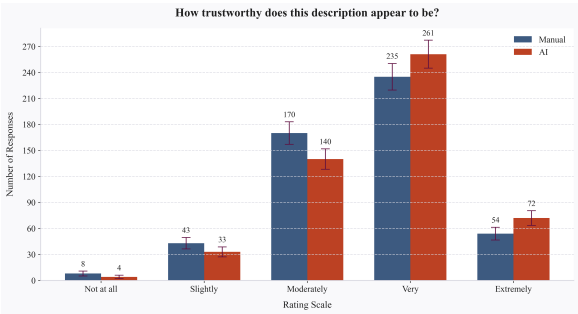

Figure 3: Comparison of perceived trustworthiness between manual and AI-generated descriptions. Error bars represent Poisson standard error estimates ($\sqrt{n}$).

4) rose from 46.1% to 51.2%. At the lower end, "Not at all" trustworthy ratings (level 1) decreased from 1.6% to 0.8%, and level 2 ratings also declined (8.4% to 6.5%). This pattern reflects a shift away from moderate trust (level 3; 33.3% to 27.4%) toward higher trust categories. A chi-square test indicated marginal statistical significance ($\chi^2(4) = 9.49$, $p = 0.050$), suggesting that AI personalization meaningfully improved perceived trustworthiness.

4.3 Clarity and Completeness

Participants evaluated how clear and complete the information was using a 5-point scale from Very Unclear to Very Clear. Figure 4 shows the distribution of clarity ratings.

AI-generated descriptions demonstrated superior clarity and completeness ratings compared to manual versions. The proportion of responses at level 4 ("Clear") increased from 51.0% (manual) to 55.7% (AI), and level 5 ("Very Clear") increased from 13.1% to 17.2%. This improvement coincided with a substantial reduction in neutral responses (level 3; 28.8% to 21.4%) and a decrease at level 2 (6.5% to 4.7%), indicating that AI adaptations helped participants form more definitive positive judgments about information quality. A chi-square test confirmed statistical significance ($\chi^2(4) = 11.47$, $p = 0.0218$), providing evidence that personalized AI descriptions significantly enhanced perceived clarity and completeness.

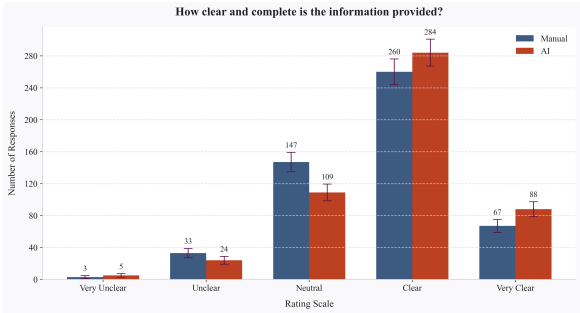

Figure 4: Comparison of perceived clarity and completeness between manual and AI-generated descriptions. Error bars represent Poisson standard error estimates ($\sqrt{n}$).

4.4 Comparative Preferences

Beyond individual ratings, participants indicated which description version they found more engaging, relevant, and appealing. For the engaging preference, responses were nearly evenly split between AI-generated (221 preferences) and manual (220 preferences), with 69 participants indicating both versions were equally engaging. This near-parity suggests that AI personalization successfully matched the engagement level of carefully crafted manual descriptions while adding the benefit of customization.

4.5 Overall Study Feedback

Following the POI evaluations, a subset of 17 participants provided overall ratings on three dimensions using a 5-point scale (1 = Poor, 5 = Excellent). The overall experience with POI descriptions received a mean rating of 3.88 (SD = 0.70, Median = 4.0), indicating generally positive reception. Participants expressed moderate to strong support for the concept of automatically adapting POI descriptions based on user interests, with a mean rating of 3.75 (SD = 0.93, Median = 4.0). Comfort with reading AI-generated descriptions when planning visits to new places received the highest mean rating of 3.94 (SD = 0.83, Median = 4.0), suggesting participants were receptive to AI-generated travel content when it demonstrated quality and relevance. The relatively low standard deviations (all below 1.0) indicate consensus among participants without extreme polarization of opinions.

5 Discussion

This study examined whether LLM-generated personalization of POI descriptions improves user engagement and perceived information quality without undermining trust. Overall, AI-adapted descriptions were rated as clearer than manual descriptions (significant improvement; $p = 0.0218$) and showed comparable trustworthiness with a marginal improvement in trust ratings ($p = 0.050$). Engagement preferences were essentially unchanged across conditions (221 AI vs. 220 manual), suggesting that personalization can enhance perceived quality without reducing willingness to engage.

The pattern of ratings indicates that personalization primarily shifts evaluations away from neutrality toward more positive judgments across significance, trustworthiness, and clarity. Trust is a

critical factor in travel decision-making, and AI-adapted descriptions received fewer "Not at all trustworthy" ratings and more "Extremely trustworthy" ratings than manual versions. A plausible mechanism is that aligning emphasis with stated interests increases perceived relevance while keeping the underlying facts stable; similarly, highlighting reader-relevant details may make the same information easier to process, yielding higher clarity without necessarily altering content.

Generalizability is limited by the sample, which was predominantly young and educated (mean age 30.94 years), with many students and engineers, and primarily Indian and German participants; responses may differ for older users, less tech-familiar groups, or other cultural contexts. For tourism platforms, the results suggest that LLMs can tailor emphasis in POI descriptions based on onboarding inputs or inferred preferences while maintaining perceived trust and improving clarity. Given the sensitivity of trust, any disclosure of AI involvement should be designed to set appropriate expectations without unnecessarily eroding credibility.

5.1 Limitations

Several limitations should be acknowledged. First, participants evaluated descriptions in a controlled study environment rather than during actual trip planning, which may not fully capture real-world decision-making contexts. Second, the study did not track whether personalized descriptions influenced actual visit intentions or behaviors beyond stated preferences. Third, the within-subjects design, while enabling direct comparisons, may have introduced demand characteristics or heightened attention to differences between versions. Fourth, the study did not examine potential negative effects of over-personalization, such as filter bubbles or reduced serendipitous discovery of POIs outside user-stated interests.

6 Future Work

Future research will first expand the study to include more diverse participants, such as older adults and people from different cultural backgrounds. This will help us understand how age and culture affect how users accept AI-generated travel content. We also plan to move beyond short-term surveys by conducting longitudinal field studies with real travel platforms. These studies will track actual user behavior, including booking rates and long-term trust. Additionally, we will test more detailed personalization by adding factors like trip purpose, travel companions, and budget constraints to see how they affect user satisfaction.

A second research direction involves the transparency and integration of these AI descriptions. We will investigate how explaining why certain information was highlighted can improve user trust. It is also important to study the potential for filter bubbles where over-personalization might limit new and surprising discoveries. We plan to develop methods that balance personalized relevance with novelty and diversity. Finally, we will explore how these adaptive descriptions can be integrated into existing recommendation systems to support users from the start of their search until they complete a booking.

7 Conclusion

This study demonstrates that Large Language Models can effectively personalize POI descriptions to improve user engagement and information quality. A study of 51 participants and 510 comparisons across 10 POIs found that AI-generated adaptations significantly improved clarity ($p = 0.0218$) and showed marginal gains in trustworthiness ($p = 0.050$). The results suggest that personalization helps users form more confident assessments by highlighting details that match their interests, with participants reporting high comfort using AI-generated content (mean = 3.94/5). These findings show that travel platforms can use LLMs to scale personalized content without the high costs of manual writing. Ultimately, this research highlights that adapting how content is described is just as important as the recommendation itself for creating a relevant and engaging user experience.

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